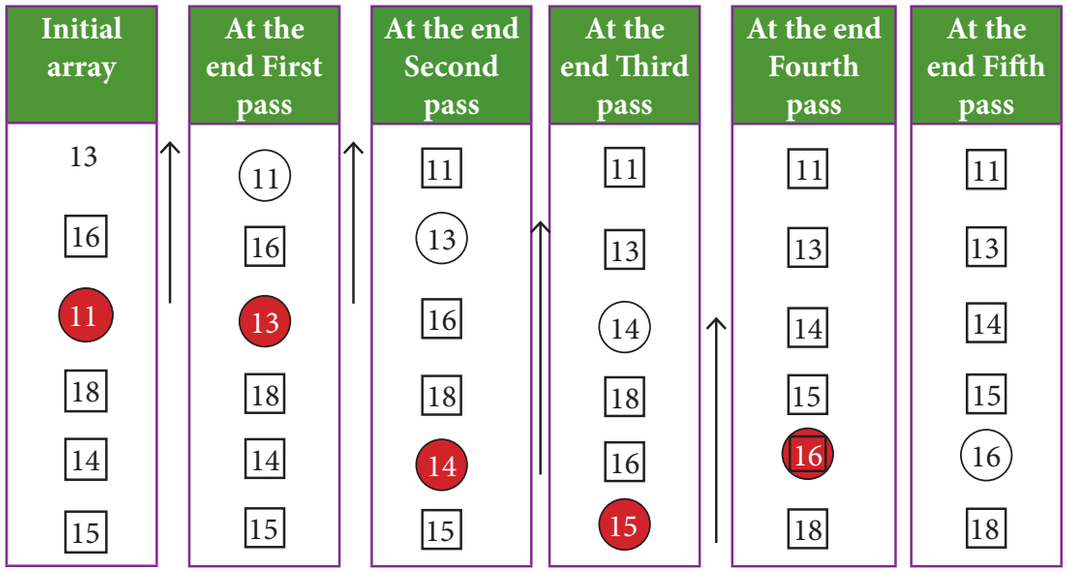
2. Now we move on to the second element position, and look for smallest element present in the sub-array, from starting index to till the last index of sub - array.
3. Now replace the second smallest identified in step-2 at the second position in the or original array, or also called first position in the sub array.
4. This is repeated, until the array is completely sorted.

Let's consider an array with values {13, 16, 11, 18, 14, 15}

Below, we have a pictorial representation of how selection sort will sort the given array.



In the first pass, the smallest element will be 11, so it will be placed at the first position.

After that, next smallest element will be searched from an array. Now we will get 13 as the smallest, so it will be then placed at the second position.

Then leaving the first element, next smallest element will be searched, from the remaining elements. We will get 13 as the smallest, so it will be then placed at the second position.

Then leaving 11 and 13 because they are at the correct position, we will search for the next smallest element from the rest of the elements and put it at third position and keep doing this until array is sorted.

Finally we will get the sorted array end of the pass as shown above diagram.

4.5.3 Insertion sort

Insertion sort is a simple sorting algorithm. It works by taking elements from the list one by one and inserting then in their correct position in to a new sorted list. This algorithm builds the final sorted array at the end. This algorithm uses n-1 number of passes to get the final sorted list as per the pervious algorithm as we have discussed.

Procedure for Insertion sort

Step 1 – If it is the first element, it is already sorted.

Step 2 – Pick next element

12th Computer Science_EM Chapter 4.indd 40 21-12-2022 15:55:01